

1. Show that under a coordinate transformation x^a to x'^m , the Christoffel symbols Γ_{jk}^i transform as:

$$\Gamma_{lp}'^m = \frac{\partial x'^m}{\partial x^i} \frac{\partial x^j}{\partial x'^n} \frac{\partial x^k}{\partial x'^p} \Gamma_{jk}^i + \frac{\partial x'^m}{\partial x^i} \frac{\partial^2 x^i}{\partial x'^n \partial x'^p}$$

2. In class, we utilized several identities to show that the four-acceleration scalar $a^\mu a_\mu$ is invariant. Prove these identities:

(a) $\gamma v = c(\gamma^2 - 1)^{\frac{1}{2}}$

(b) $c^2 d\gamma = \gamma^3 \vec{v} \cdot d\vec{v}$

3. Self study: Study the experimental set-up of Pound Rebka experiment.
4. Show that the Ricci tensor is symmetric. Can you show this choosing the locally flat coordinates?
5. Prove the Bianchi identity

$$R_{\alpha\beta\gamma\delta;\omega} + R_{\alpha\beta\delta\omega;\gamma} + R_{\alpha\beta\omega\gamma;\delta} = 0$$

6. In the class, we learnt how to construct locally flat coordinates in 4D using the local Christoffel symbols. Can you use the same construct to obtain the locally flat Euclidean coordinates on the surface of the sphere?
7. Consider a particle of mass m moving without friction and with only the constrained force on a surface of revolution $z = z(r)$ about the z-axis.

- (a) Show the square of the line-element on the surface is given by

$$ds^2 = \left[1 + \left(\frac{dz(r)}{dr} \right)^2 \right] dr^2 + r^2 d\phi^2 \quad (1)$$

- (b) Calculate the Christoffel symbols of the above surface.
- (c) What is the Riemann tensor of the surface.
8. In the class we showed that the Gaussian curvature of the 2-D sphere is given by $1/R^2$ where R is the radius of the sphere. Does the Riemann curvature give the same result as the Gaussian curvature for the 2-Sphere?

9. Define force vector as

$$F_\alpha = \frac{dP_\alpha}{d\tau} \quad (2)$$

Show that F_α under Lorentz transformation has

$$F'_0 = \gamma(F_0 - \beta F_1) \quad (3)$$

$$F'_1 = \gamma(F_1 - \beta F_0) \quad (4)$$

10. **Relativistic Ohm's law**

Show that 3-D Lorenz force

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (5)$$

can be written as

$$\frac{dP^\mu}{d\tau} = qF^{\mu\nu}v_\nu \quad (6)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and $v_\mu \rightarrow$ velocity. What is the relativistic generalization of Ohm's law ($\vec{J} = \sigma \vec{E}$)?

11. Ideal gas energy-momentum tensor: Consider a small element of an ideal gas and assume it in its rest frame, which is assumed to be locally flat, the fluid will obey the Pascal's law that the pressure is equal in all the directions.

- (a) Show that in this frame, the stress-energy tensor is diagonal. Using ideal gas equation, show that the diagonal elements are

$$T'^{\mu\nu} = \text{dia}[\rho c^2 \quad p \quad p \quad p]$$

- (b) In a general frame, metric is $g_{\mu\nu}$. Show that the stress energy tensor becomes

$$T^{\mu\nu} = (\rho c^2 + p)u^\mu u^\nu - pg^{\mu\nu}$$

Here, $g^{\mu\nu}$ is the metric in any general frame and u^μ is a 4-velocity in that frame.

12. A 2-D space has a metric given by

$$ds^2 = g_{11}(dx^1)^2 + g_{22}(dx^2)^2$$

- (a) Show that $R_{11}g_{22} = R_{22}g_{11}$.
(b) Einstein tensor is identically zero.